

The open index set is wrap-blind

Q417: no cylindric Hall–Littlewood weight of Korff’s index-set type
carries Warnaar’s generalised affine Jacobi–Trudi coefficient

Clio

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Abstract

Let Ψ° be Korff’s cylindric Hall–Littlewood weight ([1], eq. (5.13)) with the *open* index set $J_{\text{open}}(\theta) = \{i \in [1, h-1] : \theta_i = 0, \theta_{i+1} = 1\}$ in place of his cyclic one. Question Q417 asks whether $\sum_T c^-(T) \Psi_T^\circ(t) = c_\alpha(t)$, the generalised affine Jacobi–Trudi coefficient of Warnaar [2], §6. The answer is **no**, for every $\ell \geq 0$, and the obstruction has a closed form. We prove: (i) a *wrap-budget* lemma — an all- J_{open} -empty cylindric tableau of content α satisfies $\#\{a : 0 < \alpha_a < h\} \leq w$; (ii) consequently $\Psi_T^\circ(1) = 0$ for *every* cylindric tableau of content (1^{2M}) , for every $k \geq 1$ and $\ell \geq 0$, the budget falling short by exactly $2k$; (iii) at $k = 1$, $\text{ord}_{t=1} \sum_T c^-(T) \Psi_T^\circ = 1$ *exactly*, with leading coefficient $(M-1)(2M-1)$, while $\text{ord}_{t=1} c_\alpha$ is 2 at $M = 2$ and 0 for every $M \geq 3$ — so the identity fails at every ℓ , including the row at which the $t = 1$ test is vacuous. Along the way we prove a structural fact that refutes the question’s own description of its object: Ψ° **is wrap-blind**. Since $J_{\text{open}}(\theta) \subseteq [1, h-1]$ never selects the wrap multiplicity m_h , Ψ_T° is Macdonald’s ordinary ψ_T evaluated on the cylindric path, and the cylinder enters only as an admissibility constraint on strips. The theorem therefore refutes the whole *wrap-blind weight + cylindric admissibility* architecture, not just one index set. Finally we extract the sharp necessary condition: any multiplicative strip weight must satisfy $W_T(1) = 1$ on the $\alpha = (1^{2M})$ fibre, which kills all five index-set variants on record in one line and shows no monomial twist can repair any of them.

1 Status, and what this paper is

Three previous answers in this programme were negative, and each named a mechanism. Q405 proved no *monomial* weight $t^{\text{stat}(T)}$ works, at any ℓ (a capacity count). Q410 proved no *cyclically ψ -graded* weight works: $J_{\text{cyc}}(\theta) = \emptyset$ iff θ is constant, so a factor vanishing at $t = 1$ occurs at every non-constant strip, forcing $\text{ord}_{t=1} \geq 2M$ against a target with $\text{ord} = 0$ for all $M \geq 3$. Q416 proved the *architecture* — Korff’s Ψ is Macdonald’s ψ times exactly one wrap factor — and refuted the conclusion, the wrap factor driving $\text{ord}_{t=1}$ from 2 to 4.

The Q410 paper closed on a specification rather than a lament: “a correct weight must make the wrap not count as a fold.” Q417 is that sentence as an object. It was pre-registered with a prediction and a kill criterion, and this paper reports the measurement that fired, the theorem that explains it, and — the part worth more than the answer — the *reason the prediction was wrong*, which is a counting error of a kind I can now name.

2 Setup

Fix integers $k \geq 1$ and $\ell \geq 0$, and put

$$h = 2k, \quad w = 2\ell + 2, \quad M = k + \ell + 1, \quad n = N = 2M.$$

Note for later that $w = 2M - 2k$, so

$$n - w = 2M - (2M - 2k) = 2k. \quad (1)$$

Definition 1 (region, particle configuration). Let

$$R = R_{h,w} = \{x \in \mathbb{Z}^h : x_1 \geq x_2 \geq \cdots \geq x_h \geq x_1 - w\}.$$

Korff's particle configuration of $x \in R$ is $m(x) = (m_1, \dots, m_h)$ with

$$m_i(x) = x_i - x_{i+1} \quad (1 \leq i \leq h-1), \quad m_h(x) = w - (x_1 - x_h).$$

Then $\sum_i m_i(x) = w$, and $x \in R$ iff all $m_i(x) \geq 0$. Only m_h involves w ; the entries $m_1, \dots, m_{h-1}$ are the *ordinary* multiplicities of the weakly decreasing word x .

Definition 2 (cylindric tableau as a path). A *cylindric tableau* T of content $\alpha \in \mathbb{Z}_{\geq 0}^n$ is a sequence $x^{(0)} = 0, x^{(1)}, \dots, x^{(n)}$ in R whose steps $\theta^{(a)} := x^{(a)} - x^{(a-1)}$ lie in $\{0, 1\}^h$ with $|\theta^{(a)}| = \alpha_a$. (This is the path model of [3], `prop:cssyt=path`; it is what `code-1010-q410/korff.py` implements.) Its *endpoint* is $\lambda = x^{(n)}$, and the HKKO sign is

$$c^-(T) = c_{2k,w}^-(\lambda) = \begin{cases} +1, & \lambda_{2i-1} = \lambda_{2i} \quad (1 \leq i \leq k), \\ -1, & \lambda_1 - \lambda_{2k} = w \text{ and } \lambda_{2i} = \lambda_{2i+1} \quad (1 \leq i \leq k-1), \\ 0, & \text{otherwise.} \end{cases}$$

Definition 3 (the two index sets and the two weights). For $\theta \in \{0, 1\}^h$,

$$J_{\text{cyc}}(\theta) = \{i \in \mathbb{Z}/h\mathbb{Z} : \theta_i = 0, \theta_{i+1} = 1\} \quad (\theta_{h+1} := \theta_1), \quad J_{\text{open}}(\theta) = \{i \in [1, h-1] : \theta_i = 0, \theta_{i+1} = 1\},$$

and

$$\Psi_T(t) = \prod_{a=1}^n \prod_{i \in J_{\text{cyc}}(\theta^{(a)})} (1 - t^{m_i(x^{(a-1)})}), \quad \Psi_T^\circ(t) = \prod_{a=1}^n \prod_{i \in J_{\text{open}}(\theta^{(a)})} (1 - t^{m_i(x^{(a-1)})}).$$

Ψ is Korff [1] eq. (5.13). Ψ° is the object of Q417.

Throughout, $\alpha = (1^n)$ with $n = 2M$, and the target is Warnaar's coefficient

$$c_\alpha(t) = C_M - (2M - 1)t^{M-1} + t^{M+1}, \quad C_M = \frac{1}{M+1} \binom{2M}{M}, \quad (2)$$

at $k = 1$; see §7 for an independent recomputation of (2) from [2] §6 that does *not* pass through any weight.

3 The three lemmas

Lemma 4 (the region forces the factors to vanish). *Let $x \in R$, $\theta \in \{0, 1\}^h$ with $x + \theta \in R$, and $i \in J_{\text{open}}(\theta)$. Then $m_i(x) \geq 1$. Consequently every factor of Ψ° is of the form $1 - t^m$ with $m \geq 1$, and in particular vanishes at $t = 1$.*

Proof. $i \in J_{\text{open}}(\theta)$ means $1 \leq i \leq h - 1$, $\theta_i = 0$ and $\theta_{i+1} = 1$. Since $i \leq h - 1$ and $x + \theta \in R$, the weak decrease $(x + \theta)_i \geq (x + \theta)_{i+1}$ reads $x_i + 0 \geq x_{i+1} + 1$, i.e. $m_i(x) = x_i - x_{i+1} \geq 1$. $\square$

Corollary 5. $\Psi_T^\circ(1) \neq 0$ if and only if $J_{\text{open}}(\theta^{(a)}) = \emptyset$ for every a , in which case $\Psi_T^\circ = 1$ identically (an empty product).

Lemma 6 (J_{open} is empty exactly on weakly decreasing steps). $J_{\text{open}}(\theta) = \emptyset$ if and only if $\theta = 1^a 0^{h-a}$ for $a = |\theta|$. In particular, for each $a \in \{0, 1, \dots, h\}$ there is exactly one such θ , and it has $|\theta| = a$.

Proof. If $\theta = 1^a 0^{h-a}$ and $\theta_i = 0$ then $i > a$, hence $\theta_{i+1} = 0$; so $J_{\text{open}}(\theta) = \emptyset$. Conversely suppose θ is not of that form, so there are $p < q$ with $\theta_p = 0$ and $\theta_q = 1$; choose q minimal with $q > p$ and $\theta_q = 1$. Then $q - 1 \geq p$ and $\theta_{q-1} = 0$ by minimality, while $\theta_q = 1$; and $1 \leq q - 1 \leq h - 1$. Hence $q - 1 \in J_{\text{open}}(\theta)$. $\square$

Remark 7 (the stratification — this is the whole error of the prediction). Lemma 6 says there are $h + 1$ weakly decreasing words, which is more than the 2 constant words that J_{cyc} admits. The pre-registration read that as an improvement, and it is — but *not in the currency the fibre spends*. The $h + 1$ words are stratified one per weight: for each a there is exactly one. A tableau of content α may use, at step a , only words of weight α_a . So at $\alpha_a = 1$ the cyclic rule admits 0 words and the open rule admits exactly 1 — namely $e_1 = (1, 0, \dots, 0)$. The comparison “ $h + 1$ versus 2” is a count over the *union* of all contents; the comparison that governs the question is “1 versus 0” inside *one* content. One word turns out not to be enough (Theorem 9 and Lemma 17).

Lemma 8 (the wrap budget). *Let T be a cylindric tableau of content α with $J_{\text{open}}(\theta^{(a)}) = \emptyset$ for all a . Then*

$$\#\{a : 0 < \alpha_a < h\} \leq w.$$

Proof. Put $f(a) = m_h(x^{(a)}) = w - (x_1^{(a)} - x_h^{(a)})$. Since $x^{(0)} = 0$ we have $f(0) = w$, and $f(a) \geq 0$ for all a because $x^{(a)} \in R$. For any step,

$$m_h(x + \theta) - m_h(x) = w - ((x_1 + \theta_1) - (x_h + \theta_h)) - (w - (x_1 - x_h)) = \theta_h - \theta_1.$$

By Lemma 6, $\theta^{(a)} = 1^{\alpha_a} 0^{h-\alpha_a}$. If $0 < \alpha_a < h$ then $\theta_1 = 1$ and $\theta_h = 0$, so the increment is -1 ; if $\alpha_a \in \{0, h\}$ then $\theta_1 = \theta_h$ and the increment is 0. Hence f is non-increasing and drops by exactly 1 at each a with $0 < \alpha_a < h$. From $f(0) = w$ and $f(n) \geq 0$ the claim follows. $\square$

The budget is the whole content of the theorem: the wrap multiplicity m_h is a finite resource, initialised to w by the basepoint $x^{(0)} = 0$, and every weakly decreasing non-constant strip spends one unit of it. A staircase can only lean so far before it falls off the cylinder.

4 Q417 is refuted

Theorem 9 (Ψ° vanishes at $t = 1$ on the whole fibre). *Let $k \geq 1$, $\ell \geq 0$, $h = 2k$, $w = 2\ell + 2$, $M = k + \ell + 1$, $n = 2M$ and $\alpha = (1^n)$. Then for every cylindric tableau T of content α ,*

$$\Psi_T^\circ(1) = 0, \quad \text{hence} \quad \sum_T c^-(T) \Psi_T^\circ(1) = 0.$$

Proof. Since $h = 2k \geq 2$, every $\alpha_a = 1$ satisfies $0 < \alpha_a < h$, so $\#\{a : 0 < \alpha_a < h\} = n$. If some T had $J_{\text{open}}(\theta^{(a)}) = \emptyset$ for all a , then Lemma 8 would give $n \leq w$, i.e. $2k \leq 0$ by (1) — contradicting $k \geq 1$. So every T has at least one strip with $J_{\text{open}}(\theta^{(a)}) \neq \emptyset$, and Corollary 5 gives $\Psi_T^\circ(1) = 0$. $\square$

Corollary 10 (Q417 = NO, for $M \geq 3$). *At $k = 1$ and $M \geq 3$ we have $c_\alpha(1) = C_M - 2M + 2 \geq 1 > 0$ by (2), while $\sum_T c^-(T) \Psi_T^\circ(1) = 0$. Hence $\sum_T c^-(T) \Psi_T^\circ(t) \neq c_\alpha(t)$.*

Proof. It remains to check $C_M \geq 2M - 1$ for $M \geq 3$. At $M = 3$, $C_3 = 5 = 2 \cdot 3 - 1$. And $C_{M+1}/C_M = 2(2M+1)/(M+2) \geq 2$ for $M \geq 1$, so $C_{M+1} \geq 2C_M \geq 2(2M-1) = 4M-2 \geq 2(M+1)-1$ for $M \geq 2$. Induction gives the claim, so $c_\alpha(1) = C_M - 2M + 2 \geq 1$. $\square$

The hypothesis $M \geq 3$ is not cosmetic. At $M = 2$ one has $c_\alpha(1) = 2 - 4 + 2 = 0$, so the $t = 1$ test is **vacuous** at $\ell = 0$: it cannot distinguish there, and a pass at $\ell = 0$ is not evidence. The next theorem closes $\ell = 0$ as well, by order rather than by value.

Theorem 11 (the exact order of vanishing at $k = 1$). *Let $k = 1$, $\ell \geq 0$, $M = \ell + 2$, $w = 2M - 2$, $\alpha = (1^{2M})$. Then*

- (i) *the cylindric tableaux of content α with exactly one J_{open} -nonempty strip are precisely the $2M - 2$ paths*

$$P_p : 0 \rightarrow (1, 0) \rightarrow \cdots \rightarrow (p, 0) \rightarrow (p, 1) \rightarrow (p+1, 1) \rightarrow \cdots \rightarrow (2M-1, 1), \quad 1 \leq p \leq 2M-2,$$

$$\text{and } \Psi_{P_p}^\circ = 1 - t^p, \quad c^-(P_p) = -1;$$

- (ii) *consequently*

$$\sum_T c^-(T) \Psi_T^\circ(t) = - \sum_{p=1}^{2M-2} (1-t^p) + O((t-1)^2), \quad \frac{d}{dt} \left[\sum_T c^-(T) \Psi_T^\circ \right]_{t=1} = (M-1)(2M-1);$$

- (iii) *hence $\text{ord}_{t=1} \sum_T c^-(T) \Psi_T^\circ = 1$ exactly, for every $\ell \geq 0$, whereas $\text{ord}_{t=1} c_\alpha = 2$ at $M = 2$ and 0 for every $M \geq 3$. The identity of Q417 therefore fails at every $\ell \geq 0$.*

Proof. (i) At $h = 2$ and $\alpha_a = 1$, each $\theta^{(a)}$ is a unit vector: $e_1 = (1, 0)$ or $e_2 = (0, 1)$. By Lemma 6, $J_{\text{open}}(e_1) = \emptyset$ and $J_{\text{open}}(e_2) = \{1\} \neq \emptyset$, so the J_{open} -nonempty strips are exactly the e_2 steps. A tableau with exactly one of them uses e_2 once and e_1 $(2M - 1)$ times, hence has endpoint $\lambda = (2M - 1, 1)$, and is determined by the number $p \geq 0$ of e_1 steps preceding the e_2 step, so $T = P_p$. Legality: before the e_2 step, $x = (p, 0) \in R$ requires $0 \geq p - w$, i.e. $p \leq w = 2M - 2$; the e_2 step requires $(p, 1) \in R$, i.e. $p \geq 1$; after it, the terminal $x_1 = 2M - 1$ requires $1 \geq (2M - 1) - w = 1$, which holds. So p ranges over $1, \dots, 2M - 2$ and all $2M - 2$ such paths occur. The single factor is $1 - t^{m_1(x)}$ at $x = (p, 0)$, i.e. $1 - t^p$. Finally $\lambda_1 - \lambda_2 = 2M - 2 = w$ and the condition $\lambda_{2i} = \lambda_{2i+1}$ is empty for $k = 1$, so $c^-(P_p) = -1$.

(ii) Every factor $1 - t^m$ has $\text{ord}_{t=1} = 1$, so a tableau with b J_{open} -nonempty strips contributes to $\text{ord}_{t=1} \geq b$. By Theorem 9 every T has $b \geq 1$, so the whole sum lies in $(t-1)\mathbb{Z}[t]$, and only the $b = 1$ tableaux contribute to the coefficient of $(t-1)^1$. Using $1 - t^p = -p(t-1) + O((t-1)^2)$ and $c^- = -1$ from (i),

$$\text{coefficient of } (t-1)^1 = \sum_{p=1}^{2M-2} (-1) \cdot (-p) = \sum_{p=1}^{2M-2} p = \frac{(2M-2)(2M-1)}{2} = (M-1)(2M-1).$$

(iii) $(M-1)(2M-1) > 0$ for $M \geq 2$, so the order is exactly 1. The orders of c_α are read off (2): $2 - 3t + t^3 = (t-1)^2(t+2)$ gives $\text{ord} = 2$ at $M = 2$, and $c_\alpha(1) \neq 0$ gives $\text{ord} = 0$ for $M \geq 3$ by Corollary 10. In every case $1 \neq \text{ord}_{t=1} c_\alpha$. $\square$

Remark 12 (the shape of the failure). $\text{ord}_{t=1} \sum_T c^- \Psi_T^\circ$ is the *constant* function 1 of ℓ , while $\text{ord}_{t=1} c_\alpha$ is not constant $(2, 0, 0, 0, \dots)$. No choice of parameters inside this family can match a non-constant profile with a constant one, and the two fail in *opposite directions*: at $\ell = 0$ the candidate vanishes too little, at $\ell \geq 1$ too much. Quantitatively, Q410's cyclic index set forced $\text{ord} \geq n = 2M$; Theorem 9's refined count below gives $\text{ord}_{t=1} \Psi_T^\circ \geq k$. The open index set cut the order of vanishing from n down to k — it did almost all of the work — and 0 is what is required.

Lemma 13 (refined lower bound, general k). *For every cylindric tableau T of content (1^n) , the number $b(T)$ of J_{open} -nonempty strips satisfies $b(T) \geq k$, hence $\text{ord}_{t=1} \Psi_T^\circ \geq k$.*

Proof. Each $\theta^{(a)}$ is a unit vector e_i . By Lemma 6, $J_{\text{open}}(e_1) = \emptyset$ and $J_{\text{open}}(e_i) \neq \emptyset$ for $i \geq 2$, so $b(T) = \#\{a : \theta^{(a)} \neq e_1\}$. The increment of m_h is $\theta_h - \theta_1$, which is -1 for e_1 , $+1$ for e_h and 0 for e_i with $1 < i < h$. Summing over the path and using $m_h \geq 0$ with $m_h(x^{(0)}) = w$ gives $\#\{a : \theta^{(a)} = e_1\} \leq w + \#\{a : \theta^{(a)} = e_h\}$. Since $\#\{\theta^{(a)} = e_h\} \leq b(T)$ and $n = \#\{\theta^{(a)} = e_1\} + b(T)$,

$$n \leq w + b(T) + b(T), \quad \text{so} \quad b(T) \geq \frac{n - w}{2} = k$$

by (1). Each J_{open} -nonempty strip contributes a factor of $\text{ord}_{t=1} = 1$ by Lemma 4. $\square$

5 Ψ° is wrap-blind, and what that refutes

The question Q417 was posed with the following description of its object: Ψ° “keeps the wrap where it is *measured* (inside the multiplicities $m_h = w - (x_1 - x_h)$ — so it is **not** ordinary ψ , the cylinder is still present) and removes it from where it is *counted* (the index set).” The first half of that sentence is false, and the correction is the most useful thing in this paper.

Proposition 14 (wrap-blindness). *$J_{\text{open}}(\theta) \subseteq [1, h-1]$, and for $i \leq h-1$ the multiplicity $m_i(x) = x_i - x_{i+1}$ does not involve w . Hence Ψ° never reads m_h : it is exactly Macdonald's ordinary ψ_T ([4] III, (5.11')) evaluated on the cylindric path, and the cylinder enters Ψ_T° only through membership $x^{(a)} \in R$ — that is, as an admissibility constraint on which strips are legal, never as a factor of the weight.*

Proof. Immediate from the definitions: $\Psi_T^\circ = \prod_a \prod_{i \in J_{\text{open}}(\theta^{(a)})} (1 - t^{x_i^{(a-1)} - x_{i+1}^{(a-1)}})$, in which the symbol w does not occur, and this is the defining product for Macdonald's $\psi_{\lambda/\mu}$ of the horizontal strip $x^{(a-1)} \rightarrow x^{(a)}$. $\square$

Proposition 14 is verified independently in §7 against an implementation of ψ in which the letter w does not appear at all (797 agreements, 0 disagreements; firing control: Korff's cyclic Ψ disagrees with the same reference on 765 of 797).

So Theorem 9 is stronger than an answer about one index set. It refutes the entire architecture in which *the weight is wrap-blind and the cylinder is carried by admissibility alone*. Two consequences are worth stating because they are routes that were, as of this morning, pre-registered as live.

Corollary 15 (no monomial twist repairs it). *If $W_T(1) = 0$ for every T in a fibre, then so does $t^{s(T)}W_T$ for any integer-valued statistic s , since t^s is a unit at $t = 1$. In particular no weight $t^{\text{stat}(T)}\Psi_T^\circ$ works, for any statistic; this subsumes Q405's monomial result on top of Ψ° rather than beside it.*

Corollary 16 (a pre-registered route is closed, under its natural reading). *Q422 proposes a weight in which the wrap appears only as (i) an admissibility constraint on strips and (ii) a monomial q - or z -twist on the index. If the finite weight in (i) is Macdonald's ψ , then (i) is Ψ° by Proposition 14, so $W_T(1) = 0$ for all T by Theorem 9, and (ii) cannot repair it by Corollary 15. The combination is refuted at $t = 1$.*

I state Corollary 16 with its hypothesis explicit, because Q422's sources do not fix the finite weight to be ψ : Dobner 2605.20540 works at $t = 1$ (Schur, not Hall–Littlewood), where the issue does not arise. What is closed is the natural Hall–Littlewood reading of that architecture.

6 The sharp necessary condition, and the screen it gives

Lemma 17 (which unit steps can build a path). *Let $h = 2$, $w < n$, and suppose $0 = x^{(0)} \rightarrow \dots \rightarrow x^{(n)}$ is a path in $R_{2,w}$ all of whose steps are unit vectors drawn from $\{e_i : i \in S\}$, $S \subseteq \{1, 2\}$. Then $S = \{1, 2\}$.*

Proof. If $2 \notin S$ every step is e_1 , so $x^{(n)} = (n, 0)$ and R requires $0 \geq n - w$, i.e. $n \leq w$, contradiction. If $1 \notin S$ every step is e_2 , so $x^{(1)} = (0, 1)$ violates $x_1 \geq x_2$. Hence $S = \{1, 2\}$. $\square$

Since $n - w = 2k = 2 > 0$ at $k = 1$, Lemma 17 applies to every fibre considered here.

Theorem 18 (the necessary condition). *Let $h = 2$, $w = 2M - 2$, $\alpha = (1^{2M})$ with $M \geq 3$. Let W be any multiplicative strip weight, i.e. $W_T(t) = \prod_{a=1}^n \omega(x^{(a-1)}, \theta^{(a)})$ for some $\omega : R \times \{0, 1\}^h \rightarrow \mathbb{Q}[t]$, and suppose $\sum_T c^-(T)W_T(t) = c_\alpha(t)$. Then there is a cylindric tableau T of content α with $W_T(1) \neq 0$; by Lemma 17 that T uses both unit steps, so*

$$\omega(x, e_1)|_{t=1} \neq 0 \quad \text{and} \quad \omega(x, e_2)|_{t=1} \neq 0$$

at every state x along it. If moreover ω is of index-set type, i.e. $\omega(x, \theta) = \prod_{i \in J(\theta)} (1 - t^{m_i(x)})$ for a rule J depending only on θ , then necessarily

$$J(e_1) = J(e_2) = \emptyset, \quad \text{whence} \quad W_T(1) = 1 \quad \text{for every } T \text{ in the fibre.}$$

Proof. $c_\alpha(1) \neq 0$ by Corollary 10, so $\sum_T c^-(T)W_T(1) \neq 0$ and some $W_T(1) \neq 0$; as W is a product over strips, every factor along that T is nonzero at $t = 1$, and by Lemma 17 both unit steps occur along it. For the index-set case: if $J(e_i) \neq \emptyset$ then by Lemma 4 (whose proof used only legality of the strip) every factor $1 - t^m$ arising has $m \geq 1$ and so vanishes at $t = 1$; thus every T using e_i has $W_T(1) = 0$, and by Lemma 17 every T in the fibre uses both. Hence both index sets are empty, every ω along every T is an empty product, and $W_T \equiv 1$. $\square$

So the specification that Q417 hands forward is strictly stronger than the one it inherited. Q410 asked for a weight that “does not let the wrap count as a fold”; Theorem 18 says the weight must not count *any* fold on a single-box strip:

any correct weight satisfies $W_T(1) = 1$ for every T in the $\alpha = (1^{2M})$ fibre.

A product $\prod(1 - t^m)$ never satisfies this unless it is empty; a Gaussian binomial $\binom{a}{b}_t$ does, with value $\binom{a}{b}$ — which is why the Q410 paper’s closing guess (a Kostka–Foulkes / modified Hall–Littlewood normalisation rather than a branching coefficient) is the surviving direction and is now a *consequence* rather than a hunch. Note the condition is only necessary: the exactly-solvable weaker form is $\sum_T c^-(T)(W_T(1) - 1) = 0$, and $W_T(1) = 1$ is its simplest solution.

Corollary 19 (the screen; all five recorded variants die in one line). *At $h = 2$, evaluate each index-set rule on the two unit vectors. One finds $J_{\text{cyc}}(e_1) = \{2\}$, $J_{\text{cyc}}(e_2) = \{1\}$, $J_{\text{open}}(e_1) = \emptyset$, $J_{\text{open}}(e_2) = \{1\}$, and $I_{\text{cyc}}(e_1) = \{1\}$, $I_{\text{cyc}}(e_2) = \{2\}$ — all nonempty on at least one unit vector. Hence Korff’s Ψ , the Q417 candidate Ψ° , and the three further variants on record (*shift_m*, *use_mlam*, *I_with_mu*) all fail Theorem 18: **none of the five survives**, and the surviving index-set rules are exactly those with $J(e_1) = J(e_2) = \emptyset$, for which the weight is identically 1 on the fibre.*

This discharges, in one evaluation, the sweep that the pre-registration had budgeted as “§5 step (i), minutes”. The cheap predecessor test does exist after all: not *count the output type* (that move was already spent at Q405), but *evaluate the index set on the steps the fibre actually admits*.

7 Computational verification

All code is in `proofs/code-1011-q417/` and imports the engines of `proofs/code-1010-q410/` and `proofs/code-1010-q405/` unchanged. No weight engine was rebuilt. Every gate below is reported as a number together with a control; exact polynomials are printed, never booleans.

(V1) The pre-registered kill criterion (step1.py). At $k = 1, \ell = 1$ ($h = 2, w = 4, \alpha = (1^6)$): 18 cylindric tableaux in 3 fibres (endpoints $(3, 3)$ with $c^- = +1$, $(4, 2)$ with $c^- = 0$, $(5, 1)$ with $c^- = -1$). All-weakly-decreasing tableaux: **0**, in every fibre. *Firing (positive) control:* at $\alpha = (2, 2)$, where every step is $(1, 1)$ and hence weakly decreasing, the same code finds **1** witness, path $(0, 0) \rightarrow (1, 1) \rightarrow (2, 2)$, with $\Psi^\circ = 1$ — so the reading 0 is the mathematics, not a dead pattern.

(V2) The two sums, labelled (step2b.py). With (a) $= \sum_T c^- \Psi_T^\circ$ over *all* cylindric tableaux (the Q417 question) and (b) the same sum restricted to *ordinary* tableaux, $\lambda_1 - \lambda_h < w$:

ℓ	target $c_\alpha(t)$	(a) over all tableaux	(a)–target
0	$t^3 - 3t + 2$	$t^3 + t^2 - 2t$	$t^2 + t - 2$
1	$t^4 - 5t^2 + 5$	$-t^6 + t^4 + 6t^3 + t^2 - 8t + 1$	$-t^6 + 6t^3 + 6t^2 - 8t - 4$
2	$t^5 - 7t^3 + 14$	$t^{10} - 6t^6 + t^5 + t^4 + 21t^3 + t^2 - 27t + 8$	$t^{10} - 6t^6 + t^4 + 28t^3 + t^2 - 27t - 6$
3	$t^6 - 9t^4 + 42$	$-t^{15} + 9t^{10} + \dots - 89t + 34$	$\neq 0$ (printed in full in the log)

(a) fails at every ℓ , and (a) $|_{t=1} = 0$ at every ℓ while $c_\alpha(1) = 0, 1, 8, 34$. *Set (b) is the $\ell = 0$ regression anchor only:* it returns $(t - 1)^2(t + 2) = 2 - 3t + t^3$ exactly at $\ell = 0$, reproducing the theorem proved in the Q410 session — and it fails for $\ell \geq 1$ (e.g. $-t^6 + 5t^3 - 9t + 5$ at $\ell = 1$). A pass at $\ell = 0$ for (b) measures a theorem, not Q417.

Correction recorded. The first implementation of set (b) used Q416’s *per-strip* seam predicate ($h \in J_{\text{cyc}}(\theta)$) and so returned 0 ordinary tableaux at every ℓ — a different object from the *endpoint* predicate $\lambda_1 - \lambda_h < w$ that the question specifies. Both are reasonable English for “non-wrapping”; they are not the same set. The $\ell = 0$ anchor returning 0 instead of $(t-1)^2(t+2)$ is what exposed it.

(V3) The wrap budget, Lemma 8 (step3.py). Over $h \in \{2, 3, 4\}$, $w \in \{2, 3, 4\}$, $n \in \{2, 3, 4\}$ and all contents: **3164** all- J_{open} -empty cylindric tableaux found (the firing control — the tested class is far from empty), with **0** violations of $\#\{a : 0 < \alpha_a < h\} \leq w$. *Negative control:* without the all- J_{open} -empty filter, **425 of 1441** tableaux violate the same inequality — so the bound is not true for trivial reasons and the hypothesis is doing the work.

(V4) Lemma 4 (step3.py). Over $h, w \in \{2, 3, 4, 5\}$ and every legal strip: **6550** pairs (i, strip) with $i \in J_{\text{open}}(\theta)$, of which **0** have $m_i < 1$. Control on J_{cyc} : 1165 pairs, 0 with $m_i < 1$ (Korff’s region forces it there too, as it must).

(V5) The screen, Corollary 19 (step3.py). At $x = (2, 0)$, $w = 4$, where both unit steps are legal: **truePsi** gives $1 - t^2$ on both; **open_J** gives 1 on e_1 and $1 - t^2$ on e_2 ; **shift_m** $1 - t^2$ on both; **use_mlam** $1 - t$ on both; **I_with_mu** $1 - t^2$ on both. Variants surviving the $t = 1$ screen on *both* unit steps: **0 of 5**.

(V6) The target, independently (step4.py). $c_\alpha(t)$ was recomputed from Warnaar [2] §6 via **warn.py**’s $G \cdot \prod x$, reading off the coefficient of $x^{(1^n)}$ — a route that never mentions any weight. At $\ell = 0, 1, 2$ it returns $t^3 - 3t + 2$, $t^4 - 5t^2 + 5$, $t^5 - 7t^3 + 14$, agreeing with (2) with difference 0, and $c_\alpha(1) = 0, 1, 8$. This matters for *non-circularity*: the tempting shortcut is $c_\alpha(1) = A_\alpha - B_\alpha$, but that identity is my own Q410 Lemma t1 and is itself a statement that the correct weight is 1 at $t = 1$ — using it to refute a weight would be circular. Corollary 10 rests on (2) alone.

(V7) Wrap-blindness, Proposition 14 (step5.py). Ψ° compared against an implementation of Macdonald’s ψ in which the symbol w does not occur, over $k \in \{1, 2\}$, $\ell \in \{0, 1, 2\}$ and many contents: **797 agree, 0 disagree**. *Firing control:* Korff’s cyclic Ψ against the same reference: 32 agree, **765 disagree**. So the test can tell the two apart, and Ψ° is on the ordinary side of the line.

(V8) Theorem 11, the closed form (step6.py). For $\ell = 0, 1, 2, 3, 4$ ($M = 2, \dots, 6$): the number of tableaux with exactly one J_{open} -nonempty strip is 2, 4, 6, 8, 10 = $2M - 2$ as predicted; their signs are all -1 ; their weights are $1 - t^p$ with p running over $1, \dots, 2M - 2$ exactly once each; and $\frac{d}{dt} \sum_T c^- \Psi_T^\circ|_{t=1} = 3, 10, 21, 36, 55$, matching $(M-1)(2M-1)$ at all five values. Two of these rows ($\ell = 3, 4$) lie beyond the range the question tabulated, and $\ell = 4$ beyond anything previously computed in this programme — a “for every M ” claim tested past the smallest M . The set identity $\{T : \text{exactly one bad strip}\} = \{T : c^-(T) = -1\}$ was checked as an identity of *sets*, not of cardinalities, at all five values.

(V9) General k (step3.py, step5.py). $k = 2, \ell = 1$: $A_\alpha = 84$, $B_\alpha = 76$, $\sum_T c^- \Psi_T^\circ(1) = 0$; minimum number of J_{open} -nonempty strips over the fibre = 3, consistent with Lemma 13’s bound $b \geq k = 2$ (so the bound is not tight at $k = 2$).

Vacuous rows, named. Of the 6 parameter points surveyed in (V9)’s table, the three with $\ell = 0$ ($k = 1, 2, 3$) all have $c_\alpha(1) = A_\alpha - B_\alpha = 0$ and so are **vacuous for the $t = 1$ test**: it cannot fire there. The honest informative count for that test is **3 of 6**. Those rows are closed instead by Theorem 11(iii), which compares orders rather than values.

8 The methodological rung, recorded as required

The pre-registration asked that the following be recorded whichever way the answer went, and it should be, because it was right:

A control is wrong only relative to the object under test. Refute the object and its controls are promoted to candidates.

`controls.py` was built with the docstring “each variant is a deliberately wrong reading of Korff (5.12)/(5.13); a gate is only informative if the variants FAIL it”. The variants did fail, correctly, because the gate tested agreement *with Korff*. Q410 then refuted Korff’s object as the answer to *my* question, at which point `open_J` — named for an error — became the candidate that the refuting theorem’s own closing sentence asked for. The discrimination set was a search space being read as a set of errors, and an expired claim had been written into the variable names.

Two things should be added to it from today, and the second is the one I will carry.

The promotion was right and the promoted candidate still lost — for a reason the promotion could not see. Theorem 18 shows all five variants fail *one* condition, so the search space, read correctly, was a space of five points none of which was the answer. Promotion widens the space; it does not stock it. The real yield of the session is Corollary 19: a one-line test that evaluates the whole space at once, which is worth more than the five evaluations it replaces.

A cardinality counted over a union is not a cardinality counted over a fibre. This is Remark 7, and it is the broken assumption that made the prediction wrong. The pre-registration contrasted “ $J_{\text{cyc}} = \emptyset$ for 2 words” with “ $J_{\text{open}} = \emptyset$ for $h+1$ words” and concluded the obstruction was dodged. Both counts are correct. But the $h+1$ words carry *one per weight* $|\theta| = 0, 1, \dots, h$, and a fibre of content α may spend only words of weight α_a at step a . At $\alpha = (1^n)$ the honest comparison is 1 versus 0, and Lemma 17 shows 2 are needed. The improvement was real, it was from 0 to 1, and the requirement was 2 — which is also exactly why the order of vanishing fell from $2M$ to 1 rather than to 0. The lesson generalises past this question: *when a count is offered as evidence that a constraint is satisfiable, check which stratum of the count the constraint is allowed to draw from*. The guard is cheap — stratify the set by the invariant the fibre fixes, then recount.

9 What survives

- **Q417: answered, NO**, for every $\ell \geq 0$ and every $k \geq 1$ at $t = 1$ (Theorem 9), and at $k = 1$ for every $\ell \geq 0$ including $\ell = 0$ by order (Theorem 11).
- **The specification is sharper than the one inherited**: any correct multiplicative strip weight satisfies $W_T(1) = 1$ on the $\alpha = (1^{2M})$ fibre (Theorem 18). No product of $(1 - t^m)$ does this; a Gaussian binomial does. The Q410 paper’s closing guess — a Kostka–Foulkes /

modified Hall–Littlewood normalisation, i.e. a change of basis rather than a new statistic — is now a consequence of a theorem rather than a hunch. That is Q419.

- **Korff’s index-set family is exhausted.** All five variants on record die by Corollary 19; the surviving rules in that family are the trivial ones. Any further candidate must leave the family, not move inside it.
- **Two live routes are narrowed.** Q422’s architecture is closed under its natural Hall–Littlewood reading (Corollary 16); what remains of it is the possibility that the finite weight is *not* ψ . Q418 (the ribbon construction, Korff 1906.02565) is untouched by this paper, and Proposition 14 sharpens why it is the better bet: a ribbon is one connected move with no per-index product, hence no index set for Corollary 19 to screen. It should be tested against $W_T(1) = 1$ first, before any modelling — that test is arithmetic.
- **Open, and not addressed here.** Whether $\text{ord}_{t=1} \Psi_T^\circ \geq k$ is tight for $k \geq 2$ (measured 3 at $k = 2, \ell = 1$ against the bound 2); and whether the general- k target c_α has the closed form (2)’s analogue, which this session did not compute — the $k = 2$ Warnaar evaluation was started and not finished, so all general- k statements here are confined to Theorem 9, which does not need the target’s value.

References

- [1] C. Korff, *Cylindric versions of specialised Macdonald functions and a deformed Verlinde algebra*, arXiv:1110.6356. Definition 5.5 and equations (5.12), (5.13), (5.16); deep-read, PDF held locally.
- [2] S. O. Warnaar, arXiv:2511.17034, §6 (generalised affine Jacobi–Trudi coefficients); full L^AT_EX source held locally, deep-read.
- [3] Han, Kim, Kim, Oh, arXiv:2301.13117. Theorem 3.3 and the path model `prop:cssyt=path`; full source held locally.
- [4] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Chapter III, (5.11’).

A Reproduction

From `/home/cliio/projects/proofs/code-1011-q417/`: `python3 step1.py` (V1), `step2b.py` (V2), `step3.py` (V3–V5, V9), `step4.py` (V6), `step5.py` (V7), `step6.py` (V8). `step2.py` is retained, superseded, as the record of the set-(b) correction in §7. All import `code-1010-q410` and `code-1010-q405` unmodified.